Exp No: 1 DATE: 16/7/26	BASIC NETWORKING COMMANDS IN LINUX AND WINDOWS OPERATING SYSTEM

~~AIM~~ &AIM: To study and execute linux and windows networking commands

LINUX BASIC NETWORKING COMMANDS

S.NO	COMMAND	PURPOSE	EXPLANATION - WHY
1.	ping ping google o/p: ~~IPv4 Add~~: 192.168.1.0 Reply from ~~Subnet mask: 255.255.255~~	Tests connectivity between two network devices	To check whether a host is reachable and measure respon time.
2.	ifconfig o/p: eth0: inet 192.168.1.20 net	Displays and configure network interface information.	To view or troubleshoot network interface settings.
3.	mtr o/p: host Loss snt 192.168.1.10 0.0 to	Combines ping and traceroute for network diagnostics	To identify packet loss and delays at each network hop.
4.	tcpdump o/p: 12:15:10 ip 192.168.1.20 > 8.8.8.8: ICMP echo request.	Captures and analyzes network packets.	To monitor and troubleshoot network
5.	ip address show o/p: eth0: inet 192.168.1.20/24 inet6 fe80::1234	Display Ip addresses network interfaces	To verify the assigned Ip address
6.	ip address add o/p: Ipaddr 192.168.1.10/24 Added.	Assigns an Ip addresses to an interface	To configure a network interface with a new IP.
7.	ip address del o/p: Ipaddr 192.168.1.10/24 removed.	Removes an Ip addresses from an interface	To delete an unwanted or incorrect IP address.

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8	ip link set o/p: link setup	Enables or disables a network interface	To bring a network interface up or down
9	ip route add o/p: default route added via 192.168.1.1	Adds a routing entry	To define the path for sending network packets
10	ip route delete o/p: default route deleted	Deletes a routing entry	To remove an unnecessary or incorrect route.
11.	ip route get o/p: 8.8.8.8 via 192.168.1 dev eth0 src 192.168.1.20	Displays the route to a destination	To verify which path packets will follow.

WINDOWS BASIC NETWORKING COMMANDS

1.	arp -a o/p: Interface: 192.168.1.10	Displays the ARP cache.	To identify devices connected to the local network
2.	hostname o/p: DESKTOP-ABC123	Displays the computer name	To identify the current system on the network.
3.	ipconfig /all o/p: Hostname DESKTOP-ABC123 IPv4 Addr 192.168.1.10	shows complete network configuration details	To troubleshoot IP, DNS, DHCP and gateway settings
4.	nbtstat -a o/p: NetBIOS remote Machine Name Table DESKTOP <00> UNIQUE WORKGROUP <00> GROUP	Displays NetBIOS information	To troubleshoot NetBIOS name resolution problems

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11.	ip route get o/p: 8.8.8.8 via 192.168.1.1 dev eth0 src 192.168.1.20	Displays the route to a destination	To verify which path packets will follow.

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2.	hostname o/p: DESKTOP-ABC123	Displays the computer name	To identify the current system on the network.
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5	netstat o/p: Proto local Add Foreign TCP 192.168.1.10:49785 192.250.190.78:443 Establish	Display active network connections and statistics	To monitor network connections and routing information
6.	nslookup o/p: Server: dns.google Address: 8.8.8.8 Name: google.com Address: 142.250.190.78	query DNS server	To find the IP address or DNS records of a domain
7.	pathping o/p: Tracing route to google 0 192.168.1.10 1 192.168.1.1 2 142.250.190.78	Combines ping and tracert with packet loss statistics	To diagnose network latency and packet loss
8	ping o/p: pinging google.com with 32 bytes of data time: 18ms TTL=117	Tests network connectivity	To check if a remote host is reachable
9	route o/p: IPv4 route table Network dest Netmask 0.0.0.0 0.0.0.0 192.168.1.1 192.168.1.10 255.255.255.0	Displays or modifies the routing table	To manage network routing paths

RESULT: The Linux and windows networking commands were executed successfully.

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